

Program : Diploma in Textile Technology	
Course Code : 4064	Course Title: Man made fibres and Knitting
Semester : 4	Credits: No Credit
Course Category: Program core	
Periods per week: 3 (L:2, T:1, P:0)	Periods per semester: 45

Course Objectives:

- To impart knowledge on the manufacture, properties & uses of various manmade fibres.
- To familiarize knitting technology.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basics of textile fibres		Textile fibres	2
Basics of fabric manufacturing		Fabric manufacture I	3

Course Outcomes:

On completion of the course, the student will be able to:

Con	Description	Duration (Hours)	Cognitive Level
CO1	Describe the manufacture, properties & uses of manmade fibers.	12	Understanding
CO2	Explain the various Blending, Texturisation and Processing of manmade fibres in spinning, weaving.	11	Understanding
CO3	Describe the terms used in knitting, knitting elements and structures.	10	Understanding
CO4	Discuss the concept of weft and warp knitting	10	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Levels
CO1	Describe the manufacture, properties & uses of manmade fibers.		
M1.01	Illustrate the manufacture, properties & uses of polyester	4	Understanding
M1.02	Illustrate the manufacture, properties & uses of polyolefin fibers, acrylic, and Polyurethane fibers	4	Understanding
M1.03	Describe the properties and uses of Aramid, Carbon and Glass fibres.	4	Understanding.
Contents: Introduction to manmade fibers - Polymerisation- Different types of spinning technics - Importance of Drawing - Types of drawings-Introduction of PET fiber & different types - Manufacture of Terylene - Production of Ethylene glycol and D M T - Polymerization - Line Sketch- Melt spinning - Hot drawing - Dry jet wet spinning method - Flow chart for Filament and staple fiber-Physical properties - Chemical properties - Uses. Introduction to polyolefines - Poly ethylene, Poly propylene and Acrylic fibre, Poly Urethane Fibre, Method of manufacturing HDPE- Raw material - Polymerization with Ziegler type catalyst-Melt spinning and Drawing- Properties, uses - Differentiate LDPE & HDPE. Polypropylene -Manufacture- Raw material, Polymerization, Melt spinning and drawing- Properties, uses. Differentiate Acrylic and Modacrylic - Production of Orlon Fiber - Wet spinning and dry spinning - Stretching - Properties and uses of Acrylic fiber. Polyurethane Fiber - Manufacture, properties and uses of Spandex - Properties and uses of PVA, Aramid (Nomex & Kevlar), Carbon & Glass fibers			
CO2	Explain the various Blending, Texturisation and Processing of manmade fibres in spinning, weaving.		
M2.01	Explain the Blending of natural and synthetic fibres.	3	Remembering
M2.02	Understand the different process of crimping of manmade yarns.	4	Understanding

M2.03	Know the changes required for processing man-made fiber in the spinning & weaving machineries & size recipe.	4	Understanding
	Series Test - I	1	

Contents:

Blending - Importance - Characteristics of Blended material - Explain the two methods of Blending - Blow room Blending- Draw frame blending - Advantages and disadvantages. Texturisation - Objectives - Properties of texturised yarn - Types of Texturized yarns - Stretch Yarn - Modified Stretch yarn - Bulk yarn- Detailed study of following Texturization methods with sketches- False twist texturization - Stuffer Box crimping- Knife edge crimping- Air texturizing- Advantages and uses of Texturized yarn. Changes in spinning line while processing man - made fibers - Blow Room - Card - Draw frame - Simplex - Ring Frame. Size recipe for - Polyester filament yarn - Polyester / Viscose - Polyester / cotton, Control of static charges and humidity.

CO3	Describe the terms used in knitting, knitting elements and structures.		
M3.01	Definition and classification of knitting and comparison of woven and knitted fabric.	2	Remembering
M3.02	Define the various terms used in knitting	2	Remembering
M3.03	Discuss various knitting elements	3	Understanding
M3.04	Discuss Various structures and stitches in knitting	3	Understanding

Contents:

Definition of Knitting - Classification of knitting - Compare woven and knitted fabrics - Compare the process of Knitting and weaving - Define the term - Courses - Wales - Gauge - Face loop - Back loop - open loop - closed loop - Loop length - Texture - Single Jersey- Double Jersey - Study of various knitting elements - Latch needle - Bearded Needle - Compound Needle - cylinder - dial - Sinker - Cams - guides - Study of knitting action of latch needle - Define knitted structures and properties of - Plain - rib - Interlock - Tightness factor - Features and properties of - Knit stitch - Miss stitch - Tuck stitch.

CO4	Discuss the concept of weft and warp knitting.		
M4.01	Illustrate the features and types of weft knitting machines	2	Understanding
M4.02	Explain the working of weft knitting machine	3	Understanding
M4.03	Discuss the features and types of warp knitting machine.	2	Understanding
M4.04	Explain the knitting action of Raschel warp knitting machine	3	Understanding
	Series Test - II	1	

Contents:

Define weft knitting - Application of weft knitted fabrics - Passage of Yarn through single Jersey circular weft knitting machine - Comparison of single jersey and double jersey knitting - Passage of yarn through V-bed flat knitting machine - Warp knitting structure - Features of warp knitting - Application of warp knitted fabrics - knitting actions of Rachel Warp knitting machine - Comparison of Rachel and Tricot warp Knitting machines.

Text / Reference:

T/R	Book Title/Author
R1	P.W. Moncrieff, Man-made Fibres, NewnesButterworth London
R2	Usenko - Processing of Manmade fabrics – Universal Publishing corp., Mumbai -2
R3	His L. Unsinji P- Yarn Texturising Technology - Universal Publishing corp., Mumbai -2
R4	J.A.Smirfitt - An Introduction to weft knitting -Marrow publication
R5	Thomson - An Introduction to warp knitting -Marrow publication
R6	David.J.Spencer - Knitting Technology - Academic publication

Online resources

Sl.No	Website Link
1	http://www.nptel.ac.in/courses
2	http://freetextbooks.blogspot.in/2011/10/handbook-of-weaving.html
3	https://www.youtube.com/
4	http://textilelearner.blogspot.in
5	http://mytextilenotes.blogspot.in